

Lab 1: Requirements Engineering & UML Use-Case Modelling

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SECTION: 5K

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall parse structured JSON audit logs, identify and mask PII fields (Social Security Numbers, Credit Cards) before persisting records to storage.	High	Pass: Masked audit log saved with SHA-256 integrity hash. Fail: Cleartext credit card number stored in database.	Protects sensitive PII and supports GDPR compliance.
FR-002	Functional	The system shall ingest structured audit logs generated by enterprise applications.	High	Pass: A valid JSON audit log is received and recorded for processing. Fail: A valid log is rejected without an error reason.	Provides centralized collection of enterprise audit events.
FR-003	Functional	The system shall allow authorized users to search and filter historical audit records.	High	Pass: A query returns only records matching the specified filters. Fail: Records outside the selected criteria are returned.	Supports compliance investigation and audit review.
FR-004	Functional	The system shall generate an immutable audit export containing selected audit records and cryptographic integrity information.	High	Pass: Export contains selected records and a valid SHA-256 integrity hash. Fail: Export is generated without integrity information.	Provides trustworthy evidence for compliance and auditing.
FR-005	Functional	The system shall allow an authorized Security Auditor to verify the integrity of an audit export using its cryptographic hash.	Medium	Pass: Unchanged export produces a matching SHA-256 hash. Fail: Modified export produces a different hash.	Detects unauthorized modification of audit evidence.
NFR-001	Nonfunc.	Audit log query endpoints must support complex filtering across 10 million historical records in under 1 second.	High	A benchmarking test confirms the target latency and security standards under simulated peak load.	Ensures responsive querying at enterprise scale.
NFR-002	Nonfunc.	The system shall preserve successfully acknowledged audit records without data loss during normal system operation.	High	Pass: 100% of acknowledged records remain retrievable after a controlled restart. Fail: Any acknowledged record is lost.	Audit records are compliance evidence and must remain reliable.

ACTORS

1 - Enterprise Application / Log Source

Provides structured audit logs to the system for centralized processing and compliance logging.

2 - Compliance Officer

Uses the system to query audit records and generate audit exports for compliance activities.

3 - Security Auditor

Uses the system to query audit records, generate audit exports, and verify the integrity of exported audit evidence.